

Absolute-Value Threshold Preservation under Bounded Score Perturbations

Research Architecture Runtime
operator/absolute-value-threshold

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Abstract

We study $A(x) = \mathbf{1}\{|x| \geq T\}$ for $T \geq 0$ under $|x' - x| \leq \varepsilon$. Pass/fail buffers use $|x| \geq T + \varepsilon$ and $|x| + \varepsilon < T$. Lean `LEAN_FULL` on Mathlib `ℝ` (`REAL_MATHLIB`).

1 Theorems

Theorem 1 (Preservation). *If $|x' - x| \leq \varepsilon$: (1) $|x| \geq T + \varepsilon \Rightarrow A(x') = 1$; (2) $|x| + \varepsilon < T \Rightarrow A(x') = 0$.*

Theorem 2 (Sharpness). *If $\varepsilon \leq x$ and $T \leq x < T + \varepsilon$, then $x' = x - \varepsilon$ fails A .*

2 Comparison

Radial lift of Threshold; related to Sign via the origin cut. Fail side uses $|x| + \varepsilon < T$ (strict form of the half-open band).

3 Lean / certificate

`Research.Operators.AbsThreshold.Preservation`; `lean/certificates/absolute-value-threshold/abs-th`

4 Limitations

Mathlib `ℝ` (`REAL_MATHLIB`); $T \geq 0$; sharpness stated on the nonnegative ray with $\varepsilon \leq x$.